

Next Generation Networks

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Course agenda

- Three hours per week, mostly lectures, but also tutorials, tests, talks, discussions etc.
- Lecturers:
 - Marco Ruffini on the fixed/optical network part
 - Nicola Marchetti on the wireless network part
- The course marks are as follows:
 - 70 % for final exam
 - 30% for course work
 - 2x tests based on lectures + research papers + talks

Material for the optical part

1. Optical Transmission systems

- a) Propagation in fibre, dispersion and nonlinearities
- b) Optical transmitters, modulation, direct detection and coherent receivers
- c) Link design, BER curves, optical amplification, OSNR, GOSNR, physical impairments and Margins, WDM and flexgrid
- d) Machine learning use case: characterisation of optical components

2. Quantum communications networks and protocols

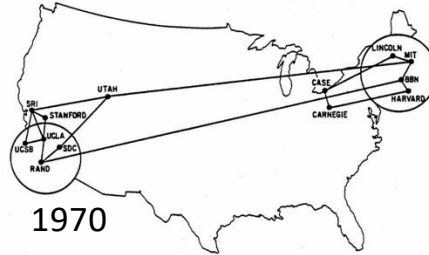
- a) Start of quantum mechanics and peculiar experiments
- b) Qubits, superposition and entanglement
- c) Simple quantum circuits
- d) Quantum key distribution
- e) Quantum networks: teleportation, entanglement swapping, entanglement distribution

Internet early evolution

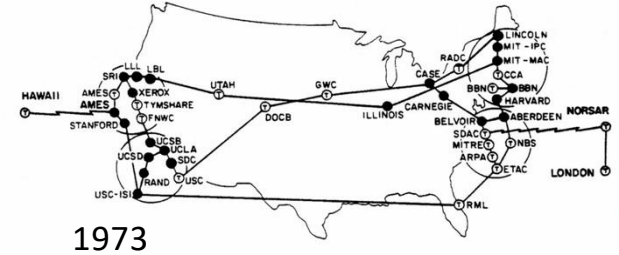
Arpanet starts...



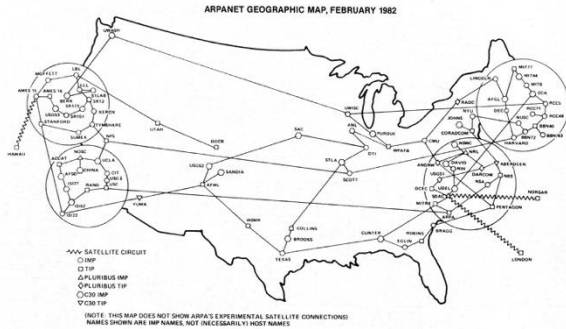
...it evolves...



...goes international...

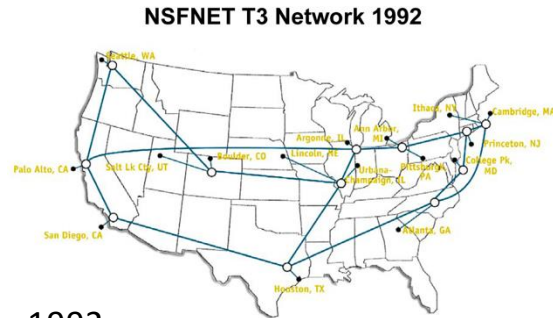


... community keeps growing...



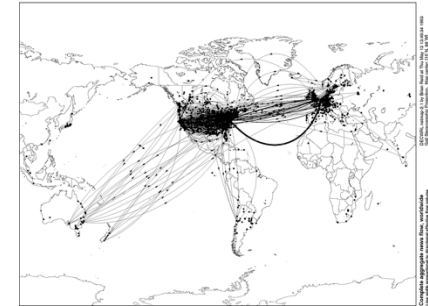
1982

... decommissioned in 1990 to be part of the Internet...



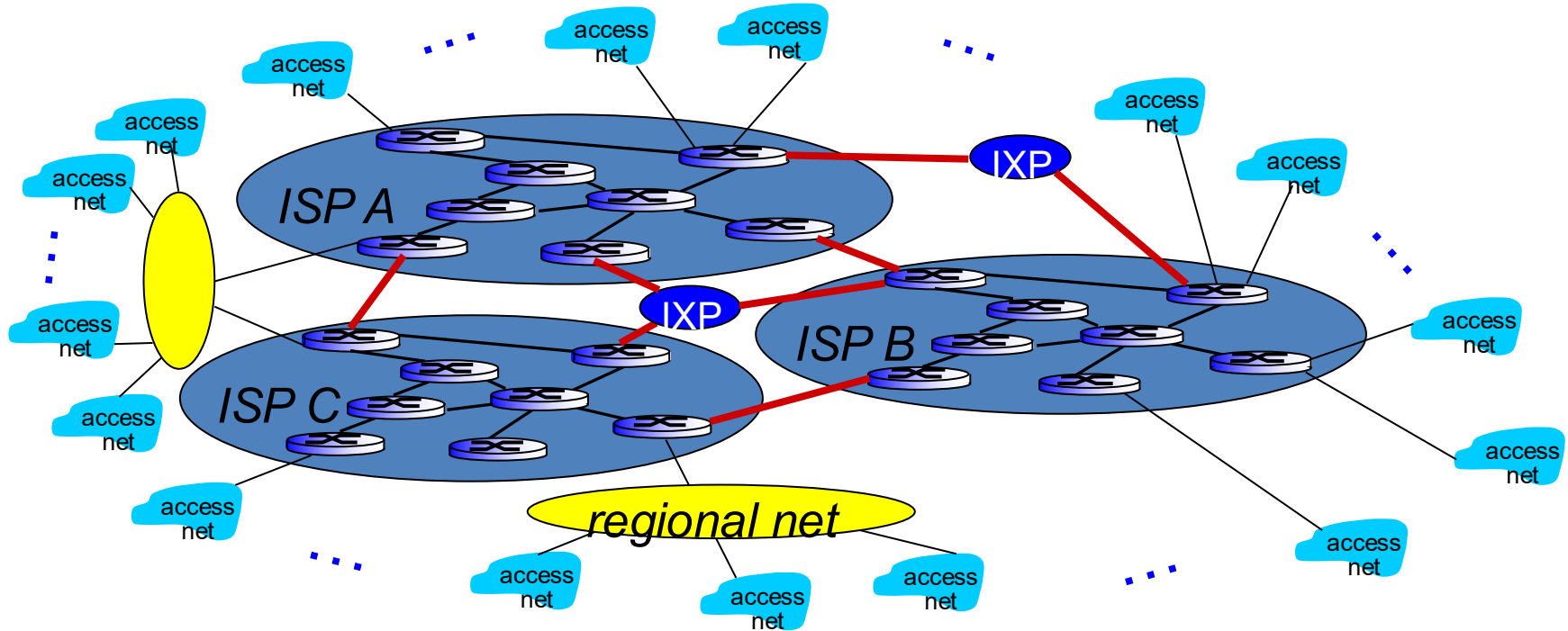
1992

...it's truly global...

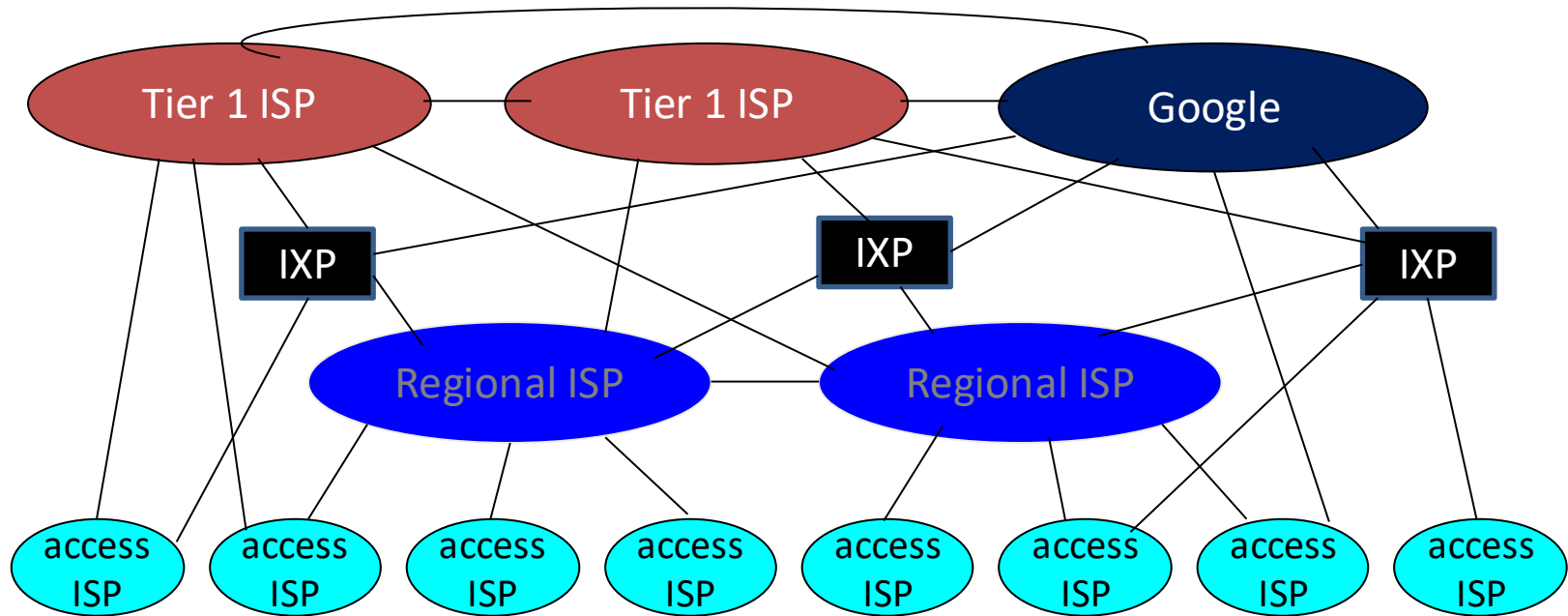


Internet structure: network of networks

... and regional networks may arise to connect access nets to ISPs



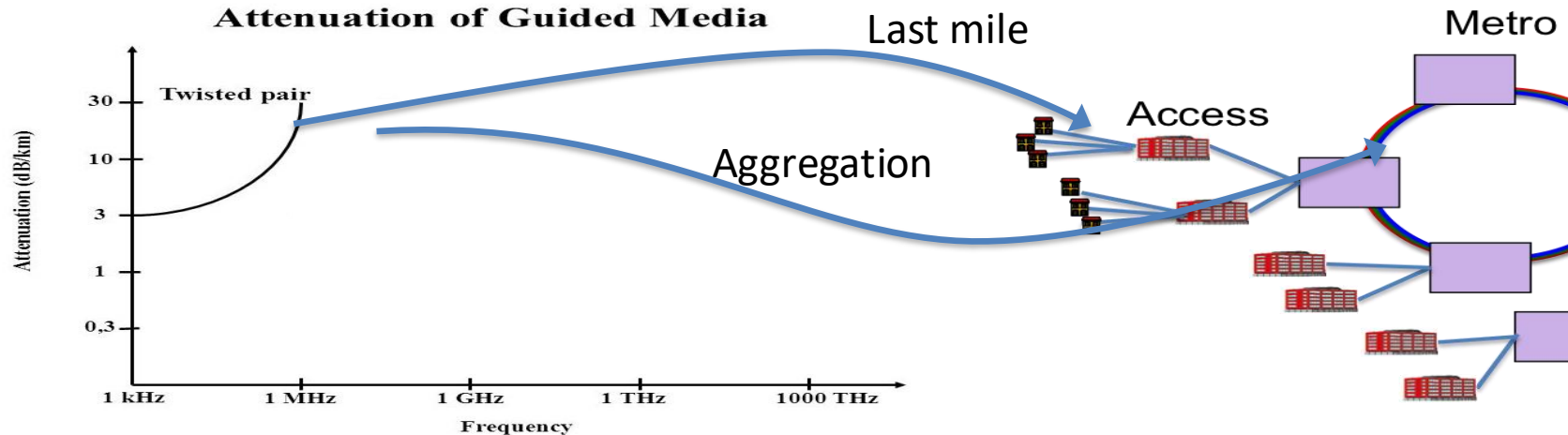
Internet structure: network of networks



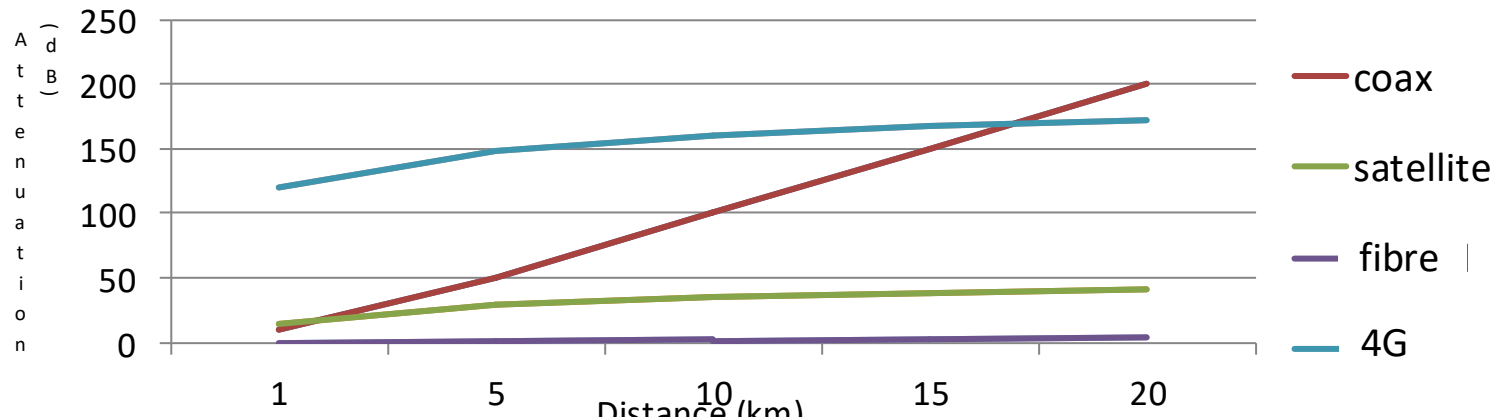
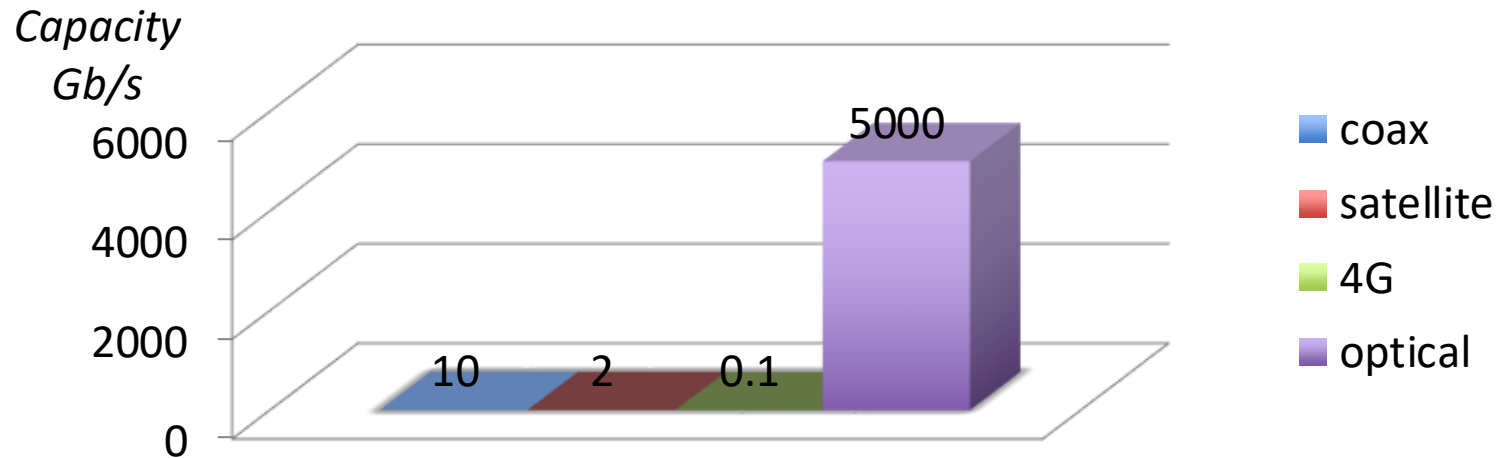
- at center: small # of well-connected large networks
 - “tier-1” commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
 - content provider network (e.g, Google): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

Transmission distances

- Telco networks were build on progressive upgrade of legacy technology
- Initially network made use of copper transmission: limited in distance to a few kilometers



Then optical communication came along



Attenuation comparison

	Frequency Range	Typical Attenuation 衰减	Typical Delay	Repeater Spacing 中继器
Twisted Pair (with loading) 铜线(双绞线)	0 to 3.5 KHz	0.2 dB/km @1kHz	50 us/km	2 km
Twisted Pair (multiple pair cable)	0 to 1 MHz	3 dB/km @1kHz	5 us/km	2 km
Coaxial Cable 同轴电缆	0 to 500 MHZ	7 dB/km @10MHz	4 us/km	1 to 9 km
Optical fiber	180 to 370 THZ	0.2 to 0.5 dB/km	5 us/km	40 km

Use of optical communications

Attenuation of Guided Media

